

Yafet Hailu

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EDUCATION

Northwestern University
B.S in **Computer Engineering**
GPA: 3.6

Evanston, IL
Graduation Date: June 2026

Relevant Courses: Computer Architecture, Operating Systems, VLSI Design, ASIC and FPGA Design, Advanced Digital Design, Real Time Digital Design and Verification with FPGAs, Parallel Computing, Microprocessor System Design, IoT Sensors and Systems

TECHNICAL SKILLS

HDL & Programming: Verilog, VHDL, SystemVerilog, HLS, C, C++, Python, RISC-V/ARM/x86 Assembly, TCL, Bash
EDA & Verification tools: Cadence (Virtuoso, Genus, Xcelium), Synopsys, Vivado, Quartus, ModelSim, SPICE, UVM, GTKWave
Platforms & Protocols: FPGA, ASIC, ARM Cortex-M, ESP32, Raspberry Pi, SPI, I2C, UART, SDR
Lab & Systems: Oscilloscope, DVM, Logic Analyzer, Xilinx / Altera FPGA, Soldering, OpenMP, GDB, MPI, Linux, Docker

PROFESSIONAL EXPERIENCES

ASIC Design Intern | *InnoMountain* May 2025 – September 2025

- Designed and implemented MMSE signal detection and channel estimation in Verilog for MIMO Wi-Fi baseband, targeting low-latency, high-accuracy decoding
- Developed and verified FPGA baseband DSP blocks (timing sync, FFT/IFFT, OFDM demodulation, spatial multiplexing) with SystemVerilog testbench, ensuring functional correctness and real-time throughput
- Verified and debugged RTL using simulation, assertions, and coverage-driven methodologies; resolved critical synthesis timing violations to achieve timing closure
- Synthesized and floor planned the baseband RTL for integration into a custom ASIC SoC, refining timing constraints, optimizing critical paths, and validating logic interoperability between digital baseband and RF interface modules

Hardware Team Lead | *Institute of Electrical and Electronics Engineers (IEEE)* Dec 2024 – May 2025

- Led the design of RGB, a piano-training platform combining real-time key recognition with LED feedback for progressive learning
- Spearheaded fullstack hardware/software co-design, integrating embedded C++ control logic with microcontroller-based sensor and LED drivers
- Built a dynamic C++ automation pipeline that parses user-selected MIDI files, extracts timing data, and drives playback synchronization for visualization

Teaching Assistant | *Northwestern University* Sep 2024 – June 2025

- Facilitated lab sessions and provided design reviews for ECE 203 (Digital Logic Design, FSMs, Assembly) and ECE 358 (Parallel Computing: Shared/Distributed Memory Architectures, OpenMP, MPI, Parallel Algorithms)
- Advised students on debugging RTL, synchronization issues, and parallel performance bottlenecks across C++ and Verilog implementations
- Supported instruction of 100+ students weekly through office hours and lab supervision, emphasizing modular, timing aware hardware design practices

SELECTED PROJECTS (More of my projects [here](#))

Open CoaXPress Deep Learning FPGA Platform — *Northwestern University* Nov 2025 – Ongoing
Capstone Project Under the supervision of Prof. Seda Öğrenci

- Building an FPGA platform supporting the CoaXPress camera interface protocol for high throughput, real-time deep learning inference.
- Using high level synthesis tools (*hls4ml*, *ml4fg*) to convert neural network models from PyTorch/TensorFlow into synthesizable hardware for deployment on FPGAs.
- Designing and integrating custom Verilog IPs for video data acquisition, preprocessing, and model acceleration.
- Developing a full FPGA-QSFP/CoaXPress pipeline to replace proprietary frame grabbers and enable flexible experimentation with advanced ML workloads.

Custom RV32IM SoC | *Verilog, FPGA, RISC-V, SoC Integration, Cadence* Dec 2024 – Present

- Designed and verified a 5-stage pipelined RISC-V CPU with hazard detection, forwarding, and branch prediction
- Synthesized the execution unit using Cadence Genus, achieving 800 MHz at 20,200 μm^2 core area
- Extended the design into an FPGA-hosted SoC, integrating UART, GPIO, timer, and memory interfaces with a bare-metal toolchain

SRAM Bank | *Cadence Virtuoso, SPICE, STA, GTK Wave* Feb 2025

- Implemented a 4x4 6T SRAM array with clocked sense amplifier, precharge, and write drivers
- Completed schematic and layout in Cadence Virtuoso; passed DRC/LVS with 100% accuracy
- Optimized cell placement for density and verified read/write timing using post-layout SPICE and Static Timing Analysis

CERTIFICATIONS AND AWARDS

Digital Design and Verification Cadence	Oct 2025
SystemVerilog Accelerated Verification with UVM Cadence	Nov 2025
Principles of Wireless Systems (5G, LTE, Wi-Fi) InnoMountain	June 2025
Linux Commands and Shell Scripting IBM	Dec 2024
Above & Beyond Computer Science (ABCS) program fellow Meta	Feb 2024